Computers in Human Behavior 73 (2017) 638–649

Contents lists available at ScienceDirect

Computers in Human Behavior

journal homepage: www.elsevier.com/locate/comphumbeh

Full length article

Expressing emotions in blogs: The role of textual paralinguistic cues in online venting and social sharing posts

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ARTICLE INFO

Article history:
Received 15 September 2016
Received in revised form
21 March 2017
Accepted 3 April 2017
Available online 10 April 2017

Keywords:
Nonverbal communication
Paralinguistic cues
Social sharing of emotion
Emotional mimicry
Venting
Social networking sites

ABSTRACT

Textual paralanguage cues (TPC) have been signaled as effective emotion transmitters online. Though several studies have investigated their properties and occurrence, there remains a gap concerning their communicative impact within specific psychological processes, such as the social sharing of emotion (SSE, Rimé, 2009). This study content-analyzed Live Journal blogposts for the occurrence of TPC in three phases of online SSE: initiation, feedback and repost. We compared these to TPC on a second type of emotional expression, emotional venting. Based on Social Information processing theory (SIP, Walther, 1992), and on the Emotional Mimicry in Context (EMC, Hess & Fischer, 2013) framework, we study predictive relationships in TPC usage in our phased model of online SSE. Results showed that TPC prevailed in SSE blogposts and strongly dominated in emotional venting posts. TPC was more common in affective feedback than cognitive. Moreover, the presence of tactile affective cues (i.e., hugs, kisses) in the initiation post predicted their presence in affective feedback. Results lend support to the idea that TPC are used in socio-contextual ways in online SSE and particularly extrapolate certain FtF nonverbal behaviors, such as the provision of socio-affective touch.

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1. Introduction

The once widespread assumption that computer mediated communication (CMC) was necessarily less emotional than face-to-face (F2F) communication has been proven wrong for about a decade (Derks, Fischer, & Bos, 2008). Textual paralanguage cues (TPC) such as emoji's or emoticons (:-D), character repetitions (*yeeei*) and nonstandard or multiple punctuations (*!!!, #%#!!*) have been found to be effective communicators of emotion in CMC (Harris & Paradice, 2007). TPC provide emotional context, strengthen the intensity of verbal messages and can be used for disambiguation (Derks, Bos, & Von Grumbkow, 2007; Riordan & Kreuz, 2010). Research has studied how receivers interpret TPC (Carr, Wohn, & Hayes, 2016; Harris & Paradice, 2007; Lo, 2008; Riordan & Trichtinger, 2017), individuals' motives to use TPC (Vandergriff, 2013; Walther & Parks, 2002, pp. 529–563), and the occurrence of TPC in different platforms (Luangrath, Peck, & Barger, 2016; Kaye, Wall, & Malone, 2016). However, there is a lack of empirical knowledge about how TPCs are used in the social sharing of emotions (SSE, Rimé, 2009; Rimé, Mesquita, Boca, & Philippot, 1991).

Studying online SSE is important, since the *process* of SSE may report benefits to both the initiator and the listener. The first, due to a temporary relief of negative emotions or a re-surge of positive emotions, and the second, due to a fostering of emotional connection and closeness between both (Rimé, 2009). TPC may make a special contribution to online SSE, because of their ability to make CMC more warm, eloquent and "human," which may ultimately affect message effectivity and interactivity in online settings.

To close this research gap, we investigate the role of TPC based on the SSE framework, because it accounts for possible beneficial and reciprocal effects between the sender and receiver. SSE has been found to occur naturally in online social networks (Rodríguez-Hidalgo, Tan, & Verlegh, 2015). To analyse TPC usage, we draw on Rodríguez-Hidalgo et al. (2015)'s conceptualization of initiation, feedback, and repost (the initiator's reaction to feedback), and

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http://dx.doi.org/10.1016/j.chb.2017.04.007
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propose hypotheses for each phase. In order to understand how TPCs are used in the different phases, we draw on Social Information Processing (SIP, Walther, 1992), a framework which may explain why and how TPCs would be used to emulate non-verbal behaviours from FtF communication online.

To empirically test differences in TPC usage, we analyse emotional venting blogposts and compare TPC occurrence in these posts to online SSE. To add robustness, we also consider TPC usage in non-SSE blogposts. Introducing and operationalizing online venting (or expressing emotion 'outbursts') may be useful in identifying venting expressions in general in SNSs, and identify whether TPC may have a stronger presence in these posts.

Because SSE involves a chain of interactions between individuals, we investigate the assumption that TPC may be mimicked. Mimicry is highly dependent on the social context of the interaction, and it likely results in an increased feeling of empathy (Hess & Fischer, 2013). Although some researchers have questioned the idea that mimicry (at least in the case of emoticons) can exist, due to the asynchronous character of CMC, which some researchers argue is less spontaneous than FtF (Derks et al., 2008), we suggest that TPC correspondence between the sender and the receiver may be expected in highly contextual scenarios, such as the provision of socio-affective feedback online.

To test our assumptions, we content-analyse a sample of blogposts in Live Journal, a platform which has been proven suitable to run empirical studies on emotions and online social networks (i.e., Gaudeul & Giannetti, 2013; Zafarani, Cole, & Liu, 2010). Another reason to choose Live Journal is its character of a blogging tool with SNS features, where users create a profile and establish a list of friend connections which can be browsed, a key characteristic of a SNSs (Boyd & Ellison, 2007). In addition, Live Journal grants a greater degree of anonymity compared to other more popular SNSs (i.e. Facebook), in which natural occurrences of online SSE already were found (Rodríguez-Hidalgo et al., 2015).

In investigating these objectives, our study contributes by a) studying the use of TPCs in three phases of online SSE, b) comparing TPC usage in different forms of emotional expressions and blogposts, c) analysing the correspondence of particular cues between sender and the receiver, and d) provide evidence as to which particular nonverbal behaviours from FtF communication may be extrapolated to CMC. Our research thus advances our understanding about the role of TPC in online emotional exchanges.

2. Theoretical framework

2.1. Textual paralanguage cues in CMC

Luangrath et al. (2016) define TPC as "written manifestations of nonverbal audible, tactile, and visual elements that supplement or replace written language and that can be expressed through words, symbols, images, punctuations, demarcations, or any combination of these elements" (p. 1). Several studies have noted that TPC are effective emotion communicators (e.g., Derks et al., 2008; Harris & Paradice, 2007; Laflen & Fiorenza, 2012). TPC have been found to have two main functions. The first is message disambiguation, or helping the receiver to interpret the meaning of the message (i.e., Riordan & Kreuz, 2010; Walther & Parks, 2002, pp. 529–563). The second is increasing message intensity, by making it seem emotionally stronger (Derks et al., 2008). Moreover, a number of studies have focussed on the functions of one particular type of TPC, and found that exclamation marks and onomatopoeic words (e.g., 'HAHAHA' for laughter) emphasize the message (Laflen & Fiorenza, 2012) and make the message more expressive (Sasamoto & Jackson, 2016). Likewise, repeated punctuations set the tone of the message (i.e. 'that would be a nice idea … wouldn't it?'(Carey, 1980).

2.1.1. TPC usage intensity

The frequency or intensity of usage of TPC reported in recent studies has ranged from .6% per word in email messages (Kalman & Gergle, 2014), to .95% per word in a German chat corpus (Vandergriff, 2013). In a study of five large corpora (including discussion forums and electronic mailing lists), Riordan and Kreuz (2010) reported a base rate of 0.47% TPC per word. This low frequency does not imply that TPCs are unimportant. For example, a single parenthesis or dot can change the meaning of a long sentence. Thompson and Foulger (1996) found that a happy emoticon at the end of an unfriendly message changed its perceived hostility. Moreover, an experiment by Walther and D'Addario (2001) found that the presence of a negative element (verbal or nonverbal) changed the message interpretation to a negative one. Regarding punctuation, Gunraj, Drumm-Hewitt, Dashow, Upadhyay, and Klin (2016) found that sentences which ended with a period in text messaging were perceived as less sincere than those without.

Though the question of whether emoticons and other paralinguistic symbols can be considered as analogue to facial expressions and other nonverbal language remains being subject of some debate (i.e., Derks et al., 2007; Walther & D'Addario, 2001), for the purposes of this study we choose to focus on the nonverbal aspects only, largely based on the premise that in FtF communication, the nonverbal language employed can easily override the meaning of a verbal message, particularly when there is conflict between the two (Burgoon, 1985; Burgoon, Buller, & Woodall, 1996; Philpott, 1983). Additionally, CMC study findings seem to sustain the notion that TPC disambiguate the emotional meaning of messages and/or increase their intensity (Derks et al., 2008; Harris & Paradice, 2007). Therefore, we define TPC intensity parsimoniously as the number of times TPC appear in messages and assume that a larger number of TPC would add up to the message's emotional intensity.

Regarding TPC intensity, few studies have focussed on a wide array of paralinguistic cues in SNSs. Luangrath et al. (2016) found that approximately 21% of tweets, 19% of Facebook posts and 31% of Instagram posts, all brand-related, contained one or more TPC. However, the authors based these frequencies in the proportion of posts where TPC occurred. While we largely base ourselves in the TPCs categorizations made by the authors, and in line with Vandergriff (2013) and Riordan and Kreuz (2010), we argue that the intensity of TPC usage should be related to the length of the post. For instance, one blogpost containing 5 words with one TPC (1 out of 5 words = 20%) has likely a higher TPC intensity than a blogpost of 40 words containing 4 TPC (4 out 40 words = 10%).

2.2. The social sharing of emotion

When a person experiences an episode that affects their emotional balance, the resulting emotions are quickly expressed to recipients close to that person, triggering a process known as the social sharing of emotions (SSE) (Luminet, Bouts, Delie, Manstead, & Rimé, 2000). SSE is an intrinsic interpersonal communicative act, which requires at least two persons who communicate, i.e.: 1) the person who experiences an emotion, feeling an urge to affiliate and express it, and 2) the recipient of the shared emotion. SSE is very prevalent in the wake of emotions; at least in 60% of cases people communicate emotions to others on the same day the episode occurs. After one week, this percentage increases to 90% (Rimé et al., 1991). Furthermore, SSE is believed to occur regardless of emotion type, gender, age, culture and level of education (Rimé, Finkenauer, Luminet, Zech, & Philippot, 1998).

We conceptualize SSE as occurring in a manner that facilitates the understanding by part of the listener, when the initiator provides a balanced explanation of the emotional episode and/or the feelings associated with it. However, the expression of emotion can also occur

in an unorganized, impulsive manner. This is commonly referred to as the emotional 'venting' of emotions (Nils & Rimé, 2012).

2.3. Emotional venting

Emotional venting implies an emotional "outburst" meant to "take the feelings off one's chest" (Nils & Rimé, 2012). Venting has been equated with catharsis, or the release of all negative emotion (Scheff, 1979). Initially it was believed that venting would be psychologically beneficial, because it would enable a person to 'let go' of the negative feelings. However, venting actually exacerbates the negative emotion (Nils & Rimé, 2012), because it increases the focus towards the negative and hinders coming to terms with the negative experience (Carver, Scheier & Weintraub, 1989). Moreover, venting diminishes the possibilities of receiving constructive feedback, due to the disregard for the listener (Nils & Rimé, 2012).

Importantly, online venting differs from the more known phenomenon of online flaming. Though online flaming has common characteristics with our definition of online venting (for instance its uninhibited character), the difference is that flaming refers mostly to an uninhibited expression of *aggression* normally directed at another (O'Sullivan & Flanagin, 2003), while in online venting, the expression of emotion is uninhibited and referring mostly to feelings about oneself or a personal situation experienced.

Since TPC increase message intensity (Derks et al., 2008), we argue that blogposts depicting emotional venting will abound with these cues, because venting is supposed to occur quite intensely (an emotional 'outburst) compared to SSE blogposts, in which the expression of emotion is more composed and comprises a rather balanced account of feelings and/or the situation which provoked the episode. We propose our first hypothesis concerning the initiator blogpost:

H1a. TPC usage will be more intense in emotional venting blogposts, compared to social sharing (SSE) blogposts.

2.4. Social Information Processing (SIP)

Social Information Processing theory (SIP, Walther, 1992) recognizes that CMC lacks the nonverbal language proper of FtF communication, but proposes in turn that individuals, motivated by a deep desire to form interpersonal relationships and affinity with others, adapt their communicative behaviours using the cues available in the CMC medium to fulfil these relational goals. SIP posits that individuals are able to form interpersonal impressions and build close personal relationships through the medium, thanks to their usage of different cues. Importantly, by 'cues' the theory refers to both language and paralanguage, including the timing aspect of messages, as well as the use of TPCs such as emoticons (Walther, 2011).

SIP posits that TPCs have metacommunicative properties and can transmit affection (Walther, 1992; Walther, Loh, & Granka, 2005) SIP suggests that users can convey affection by both verbal and nonverbal behavioural expressions in CMC, and that the social context is vitally important to emit and interpret these signals. This supports our proposition that the specific communicative context of online SSE would enable certain behaviours in individuals such as socially expressing emotions and providing socio-affective feedback online (expressing encouragement, support and warmth), a context in which we would expect individuals to use TPCs relationally and to express affiliation.

2.5. Emotional mimicry

An important aspect of nonverbal communication in FtF settings is that it can be mimicked. Emotional mimicry occurs when others' posture, gestures and facial expressions are mimicked (Hess & Fischer, 2013). The Emotion Mimicry in Context (EMC) framework postulates that mimicry is highly dependent on the social context, particularly on the type of emotional signal being mimicked and the relationship between the initiator and the observer (Fischer, Becker, & Veenstra, 2012).

Importantly, EMC sees mimicry as having an important social regulation role, that is, the mimicry of certain symbols may help to manage emotions within social relationships (Hess & Fischer, 2013). Additionally, mimicry is critically dependent on whether both parties have an initial affiliative stance towards one another. Since previous experimental studies have found that emoticons can serve the same function as actual nonverbal behaviour online (Derks et al., 2007), we propose that some gestures may be mimicked in psychological and communicative affiliative contexts online. We argue that when both counterparts are engaged in successful online SSE, first the initiator is engaged in a process of narration of an emotional experience, and second, a listener provides socio-affective feedback, acting in an affiliative stance and thus experiencing a certain degree of empathy towards the initiator.

Building on SIP, we would expect users to use TPCs to adapt to the affordances of CMC to convey affinity, empathy and/or emotional support. Findings by Liebman and Gergle (2016) support this reasoning, as they found in an experiment that the use of TPC increased the perceived affinity between two persons in an Instant Messaging (IM) conversation, while this effect disappeared when TPC were absent. These postulates and findings bring out relevant questions to the fore in an online context: could TPC be used and 'mimicked' in the context of SSE in SNSs as well? We can investigate this by studying the natural process of online SSE in its three phases.

2.6. The three phases of online SSE

Rodríguez-Hidalgo et al. (2015)'s conceptualized the communication process of online SSE in three distinguishable phases: initiation, feedback and repost, or the initiator's reaction to the feedback. A diagram depicting the main phases is displayed in Fig. 1. The typical structure of a Live Journal blogpost is similar to that of other SNSs, consisting of posts by A that can initiate a cycle, followed by responses from B (or C, D and so forth) and possible reposts by A in the same comment thread. These phases may be iterative, as both the initiator and listener can comment repeatedly in the thread.

2.6.1. First phase: online SSE initiation

SSE initiation occurs when a non-fictional, personal emotional story is narrated, or when the initiator is describing a particular emotional state, or include both altogether (the situation and the feelings). Importantly, the feelings must be experienced or felt by the initiator. SSE initiation may also occur as witnessing the emotions or events occurring to somebody else that is relevant to the initiator. In this case, the author appraises this situation as personally relevant, i.e., 'I'm feeling sad because my sister did not pass her driving exam.'

Since TPC are used to display, strengthen and disambiguate emotions (Derks et al., 2008; Harris & Paradice, 2007), we may expect TPC usage in SSE initiation blogposts to differ to that of non-SSE blogposts, because the former depict emotional experiences. We pose our second hypothesis concerning SSE initiation:

H1b. TPC intensity will be more intense in SSE blogposts, compared to non-SSE blogposts.

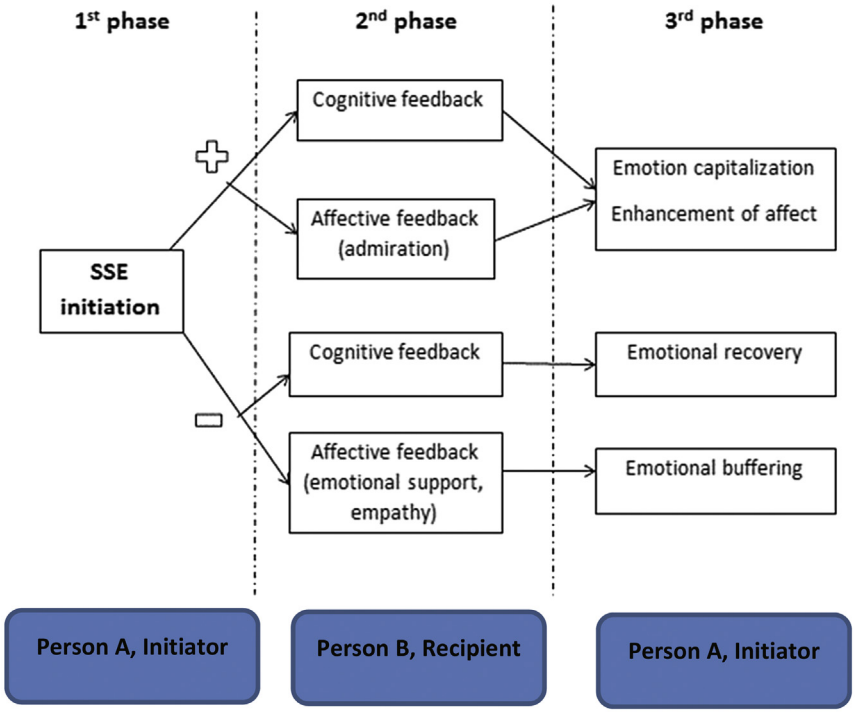


Fig. 1. The three-phased cycle of online social sharing.

2.6.2. Second phase of SSE: feedback

Feedback to SSE occurs when a person other than the initiator reacts by posting a message below the initiator blogpost. SSE theory recognizes two main types of feedback: affective and cognitive. Affective feedback appeases and emotionally contains the initiator, showing empathic or emotionally supportive behaviour. Cognitive feedback provides rationales and arguments to re-establish shattered world views, understand the event, and learn valuable lessons (Rimé, 2009). Additionally, we considered the category general feedback, which are feedback expressions which aren't affective nor cognitive.

Regarding the use of TPC, a relevant question is how these cues would be used in different types of feedback. Since TPC increase the emotional intensity of a message (Derks et al., 2008; Kavanagh, 2010), and because affective feedback has more of an emotional tone than cognitive feedback, which is more focussed on providing rationales, we build on the precepts put forward by SSE and SIP theories, to argue that users would adapt their paralinguistic expressions to the medium to provide affective feedback. We posit our first hypothesis concerning feedback:

H2. TPC usage will be more intense in affective feedback, compared to (a) cognitive feedback and (b) general feedback.

Specifically, SSE theory proposes that non-verbal haptic demonstrations of affection (i.e., hugging, kissing, a pat in the shoulder) are an important component of affective feedback in FtF communication, because they are comforting and decrease the interpersonal distance between two persons (Rimé et al., 1998). Touch has been recognized as an intensifier of emotional displays and a communicator of warmth and intimacy (Knapp & Hall, 1997). Likewise, positive expressions of touch have been related to having a prosocial disposition towards others (Hertenstein, Keltner, App, Bulleit, & Jaskolka, 2006). Extrapolating this tenet of SSE theory to online behaviour and based on SIP, we argue that there will be a higher presence of haptic affective cues (hugs, hearts, and kisses) in affective feedback, as a means to adapt to the medium and in this way, provide a feeling of being the recipient of physical affection.

H3. The usage of tactile affective paralanguage cues will be more intense in affective feedback, compared to (a) cognitive feedback, and (b) general feedback.

2.6.3. Third phase of SSE: the repost

A repost occurs when the blogpost author posts a new comment in the blogpost's comment thread, as a response to feedback received. Including reposts in our scheme allows us to investigate TPC behaviours in subsequent phases of online SSE. Since a blogpost has received feedback and reposts which occur within the specific context of SSE, and based on the Emotion Mimicry in Context theory, we can assume that at least a bulk of the nonverbal messages exchanged in the phases occur within the context of SSE. This gives us empirical ground to ask whether these cues would be mimicked, alike a FtF situation, in the comment thread. SIP theory (Walther, 1992) further supports the notion that people would take full advantage of the resources available in the medium (in this case, TPC), to adapt their nonverbal emotional expressions in contextual ways. We pose two hypotheses related to the initiation, feedback and repost phases:

H4a. The intensity of TPC in the SSE initiator blogpost will predict the intensity of TPC in feedback.

H4b. The intensity of TPC in feedback posts will predict the intensity of TPC in reposts.

3. Materials

3.1. Sample generation

The download of initiator blogposts took approximately three weeks and ended by beginning of 2014. Prior to this, permission for the download of blogposts for academic purposes was requested to

Live Journal, according to the terms of service of the platform. For this research, we made use of blogposts that were publicly available on the Live Journal platform, therefore, only blogposts that had their privacy setting set to "public" were used. Importantly, at the time of creating a Live Journal account and accepting the platform's terms of service, users declare to understand the platforms' privacy policy, in that "any personally identifiable information and/or content that you post on Live Journal that is viewable by other users may be copied and/or shared without your consent" (Live Journal Privacy Policy, 2014). The platform's terms of service further states that users who have set their privacy settings to "public" or "everybody" understand that the information may be accessed by any users of the Internet, regardless of whether they are logged in to Live Journal, and abide by the platforms' privacy policy (Live Journal terms of service, 2010). These provisions allow for the use of blogposts for academic purposes, and enable us to present a few examples to aid readers' comprehension of our coding system and results. Blogposts were downloaded according to the following criteria: a minimum length of 200 characters, having at least two comments, and belonging to a personal blog profile (thus not belonging to a certain group). The length criteria was adopted with the aim of finding complete descriptions of *emotional stories* that would qualify as SSE. The minimum of two comments was to facilitate finding 'full' cycles with feedback and reposts. In total, the download procedure generated 8.8 million posts.

Since a number of blogposts were found in other languages (Russian in particular), blogposts in English were found using Python's 'guess language' application. This reduced the sample to 7.701.284 blogposts (87.3% of the original). Importantly, from this random sample, an affect sample was generated, made with blogposts with a high presence of affect words. In order to be able to make comparisons, our study uses a proportion of blogposts taken from the random sample, and blogposts from the affect sample for the analyses. The generation of a sample with a high presence of Affect words was to increase the chances of finding blogposts where SSE took place. For further details relating to the Affect sample generation, please consult the appendix.

This study's sample consists of a randomly generated subsample from Rodríguez-Hidalgo et al. (2015)'s study. As of beginning 2017, the large majority (98%) of the blogposts in our original sample were still publicly accessible in Live Journal, which makes our sample current at the time of writing.

To be able to establish comparisons between non-SSE and the different SSE blogpost types, we drew a random sample that would yield a similar proportion between these as in our original sample. At the end of this procedure, 63 blogposts (24%) had come from the random sample and the remaining 200 (76%) from the Affect sample. Therefore, our sample had a higher prevalence of affect sample blogposts.

Blogposts were reutilized for two main reasons. Firstly, the convenience of disposing of a sample already scanned for the presence of affect words, a procedure that took several weeks to compute. Secondly, we already disposed of the human coding ratings for important variables (e.g., presence of SSE, SSE type) from our previous study.

4. Method

4.1. Coding procedure

A coding procedure to code for the presence of TPC was executed over the course of five weeks. Four English-proficient coders (different from the coders in the first study) worked in pairs (double coding). Two coders were West-European, one was South American, and the last coder was a native English speaker. To prevent possible language interpretation problems, the opinion of the native English coder was taken as the baseline in case of disagreements.

Coder training took place in three separate sessions, during which the TPC categories were explained and coders worked on practice blogposts. In subsequent sessions, disagreements and questions were resolved. After training was completed, coders were given their first batch of blogposts to code. An inter-trainer reliability test of the first 20 initiator blogposts reached a satisfactory reliability for TPC presence (Krippendorff's $\alpha = 0.89$). Afterwards, coders worked independently at home.

Coding occurred as follows: each coder pair independently coded half of the sample each, divided in two main batches of blogposts. In each batch, coder pairs coded a similar proportion of non-SSE and SSE blogposts (50/50). During the coding procedure, it was discovered that thirteen blogposts had to be removed. Six blogposts had been erased and the privacy settings of the remaining seven had been restricted.

4.2. Coding units

Coding had three main units of analysis, which can be found within one single blogpost (a blogpost is the posting of an individual text entry by the blog author in his or her blog profile). Hence, the blogpost is different than the blog profile, which constitutes a much larger unit. The three main units correspond to the three phases of online SSE: the initiator blogpost (IB), the feedback and repost. The full codebook, as well as URLs containing examples, can be obtained upon request to the first author.

4.3. Coding categories

4.3.1. Textual paralanguage cues

The TPC categories used in this study were strongly inspired on the Textual Paralanguage (TPL) typologies of Luangrath et al. (2016). However, we made two minor adaptations. The first is that the term TPL was modified to Textual Paralanguage Cues (TPC), as the word 'cue' is thoroughly employed in the CMC literature to refer to paralinguistic symbols (i.e. Walther et al., 2005). The second adaptation refers to our TPC category 'kinesics' and 'tactile,' which are outlined below. Therefore, all TPC definitions below follow Luangrath et al. (2016), unless stated otherwise.

4.3.2. TPC: auditory cues

Auditory cues indicate how the words should be spoken, conveying sound characteristics. These cues are comprised of voice quality and vocalizations. Voice qualities strengthen the intonation (pitch), pronunciation, stress (emphasis) and tempo (rhythm). They are comprised of capitalizations, underlining, punctuation and special characters, or a combination of these, for instance: 'I REALLY want that!!,' 'it's m-i-n-e,' 'Best.Joke.Ever.' '"WHAT?@#@#.' Vocalizations refer to 'utterances, fillers, terms' (Luangrath et al., 2016, p. 101) that result in audible noise of non-verbal expressions (for instance, 'uhm,' 'hmmm,' 'ugh'). Our coding scheme added onomatopoeic expressions, because of their phonetic imitation of the source of the sound (i.e., '*boom*' for an explosion). Additionally, 'yeah' was considered a voice quality (despite it being a word), since it has an utterance character.

4.3.3. TPC: tactile cues

Luangrath et al. (2016) defined 'tactile kinesics' as involving any type of physical movement with another (i.e., *high five* as involving touch between two people and the smiling emoji, which depicts a facial movement, respectively). However and to fulfil the purposes of this study, we felt that there was a need to differentiate

between affective and non-affective movements. For instance, *slap* represents a negative body movement towards another, while *dancing* may represent physical movement alone *or* with another. For this reason, our coding scheme considered tactile cues as those TPC which imply affectionate gestures towards another, for instance the words kiss, HUGS, HEART, and so forth. Given our interest to detect nonverbal signs of affection in affective feedback, we thus deviated from Luangrath et al. (2016) and considered affective tactile cues as exclusively using physical touch to express affection.

4.3.4. TPC: visual cues

Visual TPC are defined as 'non-verbal communication related to movement of any part of the body or the body as a whole' (Luangrath et al. (2016), p. 3). We reduced this definition in our codebook to only visual representations of facial or bodily movements to express emotion, such as emoji's and emoticons.

4.3.5. TPC: kinesics

We defined kinesics TPC as expressing bodily movements *to another* (for instance, a slap) or alone (i.e. rowing), with the exception of bodily movements expressing affection (i.e. a hug emoji), which belonged to the tactile affective category.

Additionally, we excluded visual artifacts, which refer to artistic qualities of the blogs' font type, colour and other stylistic elements, because their interpretation may be more ambiguous from a nonverbal and emotive point of view (i.e. a strong blue versus a light blue background).

4.3.6. Emotional venting post

A blogpost was considered to be venting when there was an obvious display of strong emotions about a real, personal circumstance or event and the description of the event or feelings was made in an evidently unorganized, impulsive and/or careless manner.

4.3.7. Non-SSE blogpost

A non-SSE blogpost contained no information about (a) an initiator's emotion, or (b) an emotional episode the initiator reports to have gone through. Hence, they aren't clearly informative about a circumstance that was consequential for the initiator's affect and the initiator does not clearly describe an emotional state as a consequence of them. Examples of non-SSE initiator blogposts are provided in sections 14.1 and 14.2 of the appendix.

4.3.8. SSE blogpost

Blogposts fitting into the main characteristics of online SSE, as defined in section 2.6.1., would qualify as a SSE blogpost. One question which may arise refers to how much of the 'bulk' of the blogpost should refer to SSE. For instance: would one emotional sentence suffice to make one post qualify as SSE? The answer is that it would suffice, however, this was only very rarely encountered. Almost with no exceptions, we found that the large majority of the post concerned the description of a narrative emotional story, while the rest often added relevant details which allowed coders to "sense" the emotional state of the initiator. Furthermore, if a blogpost contained poems, lyrics or other fictional language, this was not coded as SSE, because the reference to an explicit feeling or emotional event occurring to the initiator was lacking.

4.3.9. Feedback

Feedback was coded considering the categories affective, cognitive and general. The characteristics of affective and cognitive feedback are described in section 2.6.2. General feedback consisted of all other phrases which weren't affective nor cognitive in the comment thread. An example of a comment containing both affective and general feedback can be found in reply nr. 2, example 14.5. in the appendix: "I can't believe 2007 is already here. It went by so fast. It's crazy (general feedback). I miss you girl. You're adorable!:] (affective feedback)."

4.3.10. Reposts

Reposts occur when the initiator publishes a new comment in the comment thread.

4.4. Coding negotiations

One important negotiation concerned whether coders should code for separate phrases constituting online SSE in the initiating post, instead of coding the totality of the post. After some discussion and as stated in the previous section (SSE blogpost), since SSE is about the narration of a personal emotional story, the team felt that the totality of the initiating blogposts quite frequently accounted for the totality of the SSE, making the extraction of single sentences difficult. However, separating general versus affective and cognitive phrases was more straightforward in the feedback unit, which is why the team decided to code for these three different types of feedback in separate sentences or phrases.

A second important negotiation constituted the definition of some TPC, with voice qualities presenting the greatest challenges. For instance, whether a single question or exclamation mark should be considered TPC. After some discussion, the team decided to code from two question marks and above as TPC, since the double (or more) character of the symbol would imply a stronger intonation than a single question or exclamation mark, which may simply denote intellectual curiosity or emphasis, instead of emotion.

Another point of negotiation for voice qualities were CMC abbreviations (e.g., 'cause' or '*coz*' to signal 'because,' or 'CU' for 'see you'), since these are often meant to shorten the way a word is written, and not necessarily to alter pronunciation, a key characteristic of TPC. The team decided that these abbreviations would not be considered voice qualities because their usage may be due to simple convenience. However, a few exceptions were made with abbreviations that directly resembled a spoken utterance (i.e. LOL for 'laughing out loud'). A last negotiation for voice qualities involved the use of ellipsis, particularly ' … '. After some debate, there were included in the coding because they signals a pause in the speech and may have a strong non-verbal meaning (Vandergriff, 2013).

4.5. Coding measures

The presence of the following categories was scored *yes/no* and were taken from Rodríguez-Hidalgo et al. (2015) based on the ratings of two coders: SSE post (Cohen's Kappa = 0.84, V = 0.83), venting post (Cohen's Kappa = 8.1, V = 7.9), repost (Cohen's Kappa = 8.2, V = 7.9). Reposts were calculated for all types of blogposts. The following reliabilities were calculated based on the aggregated ratings of two coder pairs from the new round of coding for this present study (no drastic differences were observed between each of the coder pair ratings before aggregating). The presence of feedback was scored yes/no: affective (Cohen's Kappa = 0.80, V = 0.81), cognitive (Cohen's Kappa = 0.80, Cramer's V = 0.81), general (Cohen's Kappa = 0.77, V = 0.78). All reliabilities were significant at $p < .001$ level. Feedback was coded for all types of blogposts.

4.5.1. Word count

Coders annotated the total amount of words of the initiator blogpost (including title), the feedback and repost. For each feedback type, coders aggregated the word count of all the sentences in which the feedback appeared in the comment thread. Tables 1–3

display the word count per category. The coding reliabilities were as follows: Initiator blogpost Krippendorff α: .95, 95% [CI: .88, .10], affective feedback Krippendorff α: .85, 95% [CI: .73, .95], cognitive feedback, Krippendorff α: .80, 95% [CI: .65, .92], general feedback Krippendorff α: .75, 95% [CI: .61, .87], initiator repost, Krippendorff α: .88, 95% [CI: .67, 1.0].

4.5.2. TPC occurrence and intensity

Coders counted and wrote down the five TPC (voice qualities, vocalizations, tactile, kinesic and visual) occurring in each of the three coding units (initiating post, feedback, and repost). TPC intensity was calculated by dividing the average of TPC occurrence between each coder pair per unit (i.e. initiator blogpost) by the total number of words of each unit or category (i.e. initiator repost, affective feedback).

5. Data analysis

Mean TPC intensity scores ranged from .011 (TPC intensity cognitive feedback, lowest) to .075 (TPC intensity affective feedback, highest). Additionally, some TPCs presented a high amount of zeros (kinesic cues in particular). Due to this, our data showed a Poisson distribution and was heavily skewed to the right. Due to the non-normality in our data (all Shapiro-Wilk tests for TPC intensity variables were significant to the $p < .001$ level), we conducted nonparametric tests. Effect sizes were calculated and interpreted according to Rosenthal (1991) and Field (2009).

To calculate the reliability of TPC presence, we used Kalpha reliability index (Hayes & Krippendorff, 2007). Kalpha can accommodate to any number of coders as well as to different types of categories (nominal, ordinal and scalar). Because our word count scorings consisted of count data, we converted scores of each coder pair to a nominal scale to calculate Kalphas. This was as following, for initiator blogpost: 1 to 250 words (1), 251 to 500 words (2), 501 to 1000 (3), 1001 to 1500 words (4) and 1501 to 2000 (5), more than 2000 (6).

For testing predictive relationships, we conducted negative binomial regressions, because this type of regression can accommodate a Poisson distribution and a rather high presence of zeros in the data (Cameron & Trivedi, 2013). Another reason to use negative binomial regression is that the data was overdispersed, in that the variance of our dependent variables was often not identical to the mean, a critical requirement to conduct Poisson regressions (Field, 2009; Hilbe, 2007).

6. Results

6.1. Descriptive statistics

83% of SSE blogposts and 76% of non-SSE blogposts contained at

Table 1
Textual paralanguage cues (TPC) occurrence, initiator blogposts.

Blog type	Word count	Vocalization		Voice quality		Kinesic		Tactile		Visual	
	M (SD)	M (SD)	%	M (SD)	%	M (SD)	%	M (SD)	%	M (SD)	%
Venting SSE	242.7 (134.4)	1.62 (1.8)	28 (59.6)	10.62 (10.09)	47 (100)	.11 (.37)	4 (8.5)	.04 (.29)	1 (2.1)	.45 (.97)	10 (21.3)
SSE	227.78 (184.12)	1.06 (1.50)	71 (46.40)	5.34 (7.71)	(79)	.033 (.21)	4 (2.61)	.098 (.39)	(7.18)	.90 (8.0)	28 (18.30)
Non-SSE	311.30 (447.54)	1.10 (3.14)	37 (33.6)	5.30 (10.37)	34 (31)	.02 (.134)	2 (1.8)	.10 (.96)	3 (2.7)	.27 (.88)	15 (13.6)

Note: Mean (M), Standard Deviation (SD) statistics based on total number of TPC occurrence (count). All frequencies and respective percentages based on TPC occurrence in total number of blogposts Venting SSE ($n = 47$), SSE ($n = 153$), Non-SSE ($n = 110$).

Table 2
Textual paralanguage cues (TPC) occurrence, according to feedback, SSE and non-SSE blogposts.

Blogpost type	Feedback type	Word count	Vocalization		Voice quality		Kinesic		Tactile		Visual	
		M (SD)	M (SD)	%	M (SD)	%	M (SD)	%	M (SD)	%	M (SD)	%
SSE	Affective	36.50 (47.76)	.58 (1.24)	41 (33.60)	1.80 (2.77)	63 (51.6)	.025 (.20)	2 (1.64)	.607 (1.11)	43 (35.2)	5.29 (21.41)	35 (28.7)
	Cognitive	53.9 (62.65)	.077 (.274)	7 (12.06)	.846 (.98)	32 (55.2)	.00 (.00)	0 (0)	.077 (.277)	5 (8.6)	.14 (.39)	7 (12.06)
	General	165.5 (473.41)	1.65 (4.52)	79 (46.5)	3.52 (6.82)	115 (67.6)	.071 (.37)	10 (5.8)	.17 (.69)	16 (9.4)	.918 (2.56)	62 (36.5)
Non-SSE	Affective	48.60 (65.06)	.55 (1.19)	6 (30)	2.15 (2.62)	15 (75)	.05 (22)	1 (5)	.50 (1.19)	5 (25)	.70 (1.26)	8 (40)
	Cognitive	62.80 (80.26)	0 (0)	0 (0)	0 (.0)	0	0 (0)	0 (0)	.20 (.44)	1 (20)	0 (0)	0 (0)
	General	202.70 (568.53)	1.91 (4.73)	55 (50)	1.91 (4.73)	74 (67.2)	0 (0)	4 (3.6)	.18 (.52)	3 (2.7)	1.22 (3.44)	40 (36.3)

Note: Mean (M), Standard Deviation (SD) statistics based on total number of TPC occurrence (count). All frequencies and respective percentages based on TPC occurrence in total number of blogposts. SSE blogposts with affective feedback ($n = 122$), cognitive feedback ($n = 57$) and general feedback ($n = 170$). Non-SSE blogposts with affective feedback ($n = 20$), cognitive feedback ($n = 5$) and general feedback ($n = 110$).

Table 3
Textual paralanguage cues (TPC) occurrence in reposts, according to SSE, non-SSE blogpost and feedback type.

Blogpost type	Feedback type	Word count	Vocalization		Voice quality		Kinesic		Tactile		Visual	
		M (SD)	M (SD)	%	M (SD)	%	M (SD)	%	M (SD)	%	M (SD)	%
SSE	Affective	77.49 (119.0)	1.69 (5.16)	24 (39.3)	2.79 (4.92)	37 (61)	.02 (.13)	1 (1.6)	.54 (1.88)	14 (23)	.44 (.84)	18 (30)
	Cognitive	53.37 (70.28)	.17 (.46)	4 (13.3)	1.23 (2.31)	17 (57)	0 (0)	0 (0)	.13 (.57)	2 (6.7)	.40 (.85)	8 (26.6)
	General	87.78 (151.2)	1.00 (3.02)	50 (40.3)	2.69 (5.88)	79 (63.7)	.02 (.12)	2 (1.61)	.14 (.70)	7 (5.6)	.48 (.90)	37 (29.8)
Non-SSE	Affective	48.60 (65.06)	.55 (1.19)	6 (30)	2.15 (2.62)	15 (75)	.05 (.22)	1 (5)	.50 (1.19)	5 (25)	.07 (1.26)	8 (40)
	Cognitive	62.80 (80.26)	0 (0)	0 (0)	1.67 (1.52)	2 (66.6)	0 (0)	0 (0)	.33 (.57)	1 (33.3)	0 (0)	0 (0)
	General	132 (481.7)	1.15 (2.68)	31 (42.5)	3.63 (13.46)	46 (63.01)	.01 (.11)	1 (1.4)	.21 (.78)	9 (12.3)	.90 (2.8)	23 (31.51)

Note: Mean (M), Standard Deviation (SD) statistics based on total number of TPC occurrence (count). All frequencies and respective percentages based on TPC occurrence in total number of SSE blogposts with reposts which received affective feedback ($n = 61$), to cognitive feedback ($n = 30$), to general feedback ($n = 124$), and in total number of non-SSE blogposts with reposts which received affective feedback ($n = 20$), cognitive feedback (($n = 3$) and general feedback ($n = 73$).

least one type of TPC. Auditory cues (vocalizations and voice qualities) were the most common (in both non-SSE and SSE blogposts), followed by visual cues, while tactile and kinesic cues appeared less frequently. Examples of non-SSE and SSE initiator blogposts can be found in the Appendix. Tables 1–3 display the occurrence and percentages of TPC in initiator blogposts, feedback and reposts.

With regards to TPC occurrence in feedback to SSE blogposts, 75% of general feedback, 77% of affective feedback and 60% of cognitive feedback contained at least one type of TPC. Auditory cues appeared much less frequently in cognitive feedback (12%) than in affective and general feedback (34%, $z = -3.709$, $p < .001$, and 47%, $z = -2.142$, $p < .05$, respectively). One example of an affective feedback sentence with auditory cues is: *"awww!! poor baby!"* in response to the initiator telling that they have the flu.

As for visual cues, these also appeared less in cognitive feedback compared to affective feedback and general feedback (12% versus 28% in affective feedback, $z = -5.851$, $p < .001$, versus 37% in general feedback, $z = -4.833$, $p < .001$). One example of a cognitive feedback sentence with visual cues is: "*just know that you can't see it all in one trip, so try to enjoy what you can and don't rush to get it all in, and most importantly, have tons of fun :-)*" Finally, tactile cues were much more frequent in affective than in cognitive and general feedback (43% versus 5%, $z = -6.609$, $p < .001$, and 43% versus 16%, $z = -3.902$, $p < .001$, respectively). One example of affective feedback with tactile cues is *"you're the best thing that has happened to me this year <3."* These results signal that TPC were generally less present in cognitive feedback both in non-SSE and SSE blogposts, while they were much more dominant in affective feedback.

With regards to reposts, 77% contained at least one type of TPC. Since there was a very low frequency of non-SSE blogposts which received cognitive feedback, the presence of TPC cues is uninterpretable. However and in spite of this, we note that the low percentage of occurrence of reposts as a response to cognitive comments in total sample composed of 103 blogposts is already noteworthy. Additionally, the presence of TPC cues is remarkably low in the few blogposts found with TPCs (3), in spite of its average word count (62 words).

6.2. Hypothesis testing

6.2.1. Initiator repost hypotheses

Hypothesis 1a predicted that TPC intensity would be higher in emotional venting blogposts, compared to social sharing (SSE) blogposts. TPC intensity of SSE venting blogposts ($Mdn = 0.048$) differed significantly from non-venting SSE blogposts ($Mdn = 0.014$), $U = 2402.50$, $z = -5.67$, $p < .001$, which represents a medium effect ($r = 0.34$). We find that TPC intensity is significantly higher in SSE venting blogposts, compared to non-venting SSE blogposts. This result suggests that TPC are significantly more present in venting posts than in non-venting emotional posts, bringing support to hypothesis 1a. As for examples, section 14.3. in the appendix displays a venting blogpost with the full cycle of SSE. In addition, Table 1 presents an overview of the frequencies of the different types of TPCs in the different types of SSE initiation posts.

Hypothesis 1b predicted that TPC intensity would be higher in social sharing blogposts (SSE blogposts), compared to non-SSE blogposts. Results show that the TPC intensity of SSE blogposts ($Mdn = 0.020$) significantly differed from non-SSE blogposts ($Mdn = 0.013$), $U = 6880.50$, $z = -2.53$, $p < .05$, ($p = 0.011$). We find that TPC intensity is significantly higher in SSE blogposts compared to non-SSE blogposts. This supports hypothesis 1b, however we note that this represents a small effect ($r = 0.015$). For instance, we can observe in example 14.1. of a non-SSE initiator blogpost in the appendix contains only one visual TPC, while example 14.5. depicts a SSE initiator blogpost with many TPCs.

6.2.2. Feedback hypotheses

Hypothesis 2a stated that the intensity of TPC would be higher in affective feedback, compared to cognitive feedback. The TPC intensity of affective feedback ($M = 0.075$, $SD = 0.165$, $Mdn = 0.126$), significantly differed from that of cognitive feedback ($M = 0.011$, $SD = 0.049$, $Mdn = 0.014$), $W = 381.500$, $z = -2.47$, $p < .050$, ($p = 0.013$), which represents a small effect ($r = 0.25$). This result suggests that TPCs are significantly more used in affective feedback compared to cognitive, which brings support to hypothesis 2a. Examples of both affective and cognitive feedback can be found in section 14.3. i.e., *"I love you hun"* for affective feedback, and *"& soon, those guys that hurt you, will realize what they missed out on"* for cognitive feedback. Table 2 presents an overview of the frequencies and intensities of the different types of TPCs in the different types of feedback.

Hypothesis 2b predicted that the intensity of TPC would be higher in affective feedback, compared to general feedback. The TPC intensity of affective feedback ($M = 0.075$, $SD = 0.165$, $Mdn = 0.126$), significantly differed from that of general feedback ($M = 0.040$, $SD = 0.170$, $Mdn = 0.068$), $W = 4210.00$, $z = -2.91$, $p < .010$, ($p = 0.004$). This result suggests that TPCs were significantly more used in affective feedback compared to general feedback, which represents a small effect ($r = 0.24$). Hypothesis 2b is supported. An example that distinguishes general from affective feedback within the same reply can be found in section 14.5. of the appendix, reply nr. 2: "*I can't believe 2oo7 is already here. It went by so fast. It's crazy," for general feedback, and "I miss you girl. You are adorable!:],*" for affective feedback.

Hypothesis 3a predicted that the intensity of tactile affective cues would be higher in affective feedback than in cognitive feedback. The intensity of tactile cues in affective feedback ($M = 0.104$, $SD = 0.141$, $Mdn = 0.064$), did not significantly differ from that of cognitive feedback ($M = 0.037$, $SD = 0.021$, $Mdn = 0.037$), $W = 494.00$, $z = -1.54$, $p = 122$). However, we note that tactile affective cues only occurred in five blogposts for cognitive feedback, due to which this result may not be interpretable due to a small sample size.

Hypothesis 3b posited that the intensity of tactile affective cues would be more intense in affective feedback compared to general feedback. The intensity of tactile affective cues in affective feedback ($M = 0.104$, $SD = 0.141$, $Mdn = 0.064$), significantly differed from that of general feedback ($M = 0.028$, $SD = 0.033$, $Mdn = 0.013$), $W = 4470.50$, $z = -2.04$, $p < .050$, ($p = 0.041$). This result suggests that tactile affective cues were significantly more used in affective feedback compared to general feedback, which represents a small effect ($r = 0.18$), which brings support to hypothesis 2b. This means that tactile cues such as *hugs* may often accompany affective phrases, such as the example taken from an affective reply in our sample: "*(((hugs))) I'm sorry you've had a hard time!*."

6.2.3. Repost hypotheses

Hypothesis 3a predicted that the intensity of TPC cues in the initiator blogpost (IB) would predict their intensity in feedback posts. Results showed that only the presence of tactile cues was positively associated with their intensity in affective feedback posts ($B = 0.770$, $p < .05$), all other cues *ns*. For cognitive and general feedback, no presence of cues in the IB were significantly associated with their presence in feedback posts. Table 4 presents the results of the negative binomial regression. Hypothesis 3a is partially supported. Example *14.5.* in the appendix depicts a full cycle blogpost where TPC mimicry is observed for the heart TPC "<3" between the initiating post and reply number 1.

Hypothesis 3b predicted that the intensity of TPC cues in

Table 4
Negative binomial regression results predicting total intensity of TPC cues, affective feedback.

	B	SE	Wald X^2	df	95% CI Exp (B)			95% CI Exp (B)	
					LL	UL		LL	UL
Cues									
Vocalizations	−0.03	.030	.007	1	−0.86	.033	.997	.920	1.079
Voice qualities	−0.27	.040	.764	1	−0.83	.076	.974	.917	1.034
Tactile	.770	.083	84.72*	1	.606	.934	2.15	1.83	2.544
Kinesic	−.161	.097	2.73	1	−3.53	.030	.851	.703	1.030
Visual textual	−.169	.082	4.24	1	−3.30	−.008	.845	.719	.992
Total cues	.044	.030	2.01	1	−0.17	.104	.156	.983	1.110

*Note. CI = confidence interval; LL = lower limit; UL = upper limit.
*p < .001.

feedback posts would predict TPC intensity in reposts. Results showed that no TPC had a significant relationship with their subsequent presence in reposts. We found no support for hypothesis 3b.

7. Discussion

This study focussed on the role of textual paralanguage cues (TPC) in online emotional expressions by content analysing a randomly generated sample of blogposts in Live Journal. Specifically, we assessed the usage intensity of five particular types of TPC within each of the three communication phases of SSE (initiator, feedback and repost), and further investigated predictive relationships between the TPC usage in each of the three phases. We find that TPC are used in socially-contextual ways in Live Journal, which conveys a relevant role to TPC, particularly in the feedback phase of online SSE to express physical affection.

7.1. TPC occurrence

Our study particularly contributes to research in investigating different types of TPC, as many studies have focussed only on one symbol, mainly the emoticon (e.g., Derks et al., 2007; Walther & D'Addario, 2001). Regarding TPC presence, we find that auditory cues generally form the bulk of TPC usage, in all three phases. This finding may seem surprising given the amplitude of research focussing on emoticons as main nonverbal cue (e.g., Kaye et al., 2016; Lo, 2008). Although previous studies have suggested the importance of altered punctuations and other creative uses of paralanguage in the expression of emotions (e.g., Hancock, Landrigan, & Silver, 2007), our finding provides further evidence that vocalizations and voice qualities may be more suggestive of the emotional tone of a message than expected, that is in an online textual platform. These textual approximations of voice qualities may deserve greater attention. This result is corroborated by Luangrath et al. (2016), who also found a prominence of auditory cues in brand messages in social media. Moreover, this finding lends support to several studies on nonverbal behaviour in CMC, which account TPC as carrying for meaningful message aspects such as tone, volume and speech rhythm (e.g., Carey, 1980).

The fact that emoticons came overall in a second place compared to voice qualities, may be a sign that this particular cue serves more to complement the message, rather than constituting it, an idea backed up by previous findings (Derks et al., 2007; Walther & D'Addario, 2001). It might be worthwhile to investigate also investigate this result may have been due to the affordances of the particular platform. For instance, Kaye et al. (2016) found higher emoticon usage in SMS messages compared to SNSs, which may be due to emoticons and emoji's being more accessible and available in SMSs.

Moreover, the rather modest presence of haptic and kinesic cues (with the exception of affective feedback), may be as well surprising, but these again may have been due to the particular affordances of the platform. An interesting avenue for research would be to investigate the use of memes or (moving) stickers to add to the TPC of kinesics to transmit nonverbal information in particular contexts in other SNSs platforms.

7.2. SSE versus non-SSE posts

Our results provide evidence that TPC were more common in messages containing SSE, compared to those messages which did not convey SSE. This finding is in line with previous research pointing at that the higher the message cues, the more that the message is emotional (Harris & Paradice, 2007) and that TPCs are used to convey emotions (Derks et al., 2008). It is important to note that the presence of TPC in non-SSE posts is still relevant in our sample, since not every non-SSE post was necessarily 'non-emotional,' because these blogposts frequently depicted songs, lyrics or narrative works of fanfiction. This highlights the importance of comparing TPC usage between SSE initiation versus purely informative texts (such as news), in order to establish more differences with purely emotional texts.

In spite of this, our result that TPC usage in SSE blogposts was more intense than in non-SSE blogposts lends support to both the SSE and SIP frameworks. The first, in that sharing emotions involved using a complete register of emotional expressions and narrations, as can be expected of the paradigm's assumption that emotions are for sharing (Rimé, 2009). In this respect, the results add depth to our previous work showing that the full cycle of socially sharing emotions occurs online (Hidalgo et al., 2015). The second theory, SIP, is supported in that users adopt the available affordances of CMC to narrate the emotional episode and/or the feelings associated with them.

7.3. Emotional venting

Our study also contributed to the literature by introducing a definition and operationalization of online venting, that is, the expression of emotions in an impulsive and inhibited manner, referring mostly to feelings about oneself or a personal situation, which lacks the more calmed explanations and story rationale of social sharing. Interestingly, our study found that these posts fully abounded with voice qualities. One practical application of our online venting definition and evidence, is to create possible online interventions to track and support individuals who may be undergoing difficult life circumstances, or are feeling emotionally unstable.

7.4. TPC in feedback

Since there is a lack of knowledge as to the particular ways people communicate nonverbally online (Riordan & Kreuz, 2010), our study presents evidence supporting the idea that TPC are used in socio-contextual ways, contingent upon the framework of the interaction. The finding that TPC were more dominant in affective feedback suggests that TPC do not only express emotions or disambiguate messages, but may also be used to express affection and emotional support, at least in the context of online SSE. This is in line with recent studies which focus on paralinguistic affordances as a means of social support (Carr et al., 2016). Moreover, this finding suggests a beneficial process of online SSE, as listeners seem ready to engage in reciprocal affective behaviour and emotionally support the individual, extending our previous work related to the

presence of empathy and emotional support in feedback (Rodríguez-Hidalgo et al., 2015). It may be worth to investigate other forms of paralinguistic emotional support in newer platforms (i.e. Instagram, Snapchat) in line with Andalibi, Ozturk, and Forte (2017) which found evidence of affective social support in Instagram in the case of depression.

7.5. Tactile affective cues in feedback

Our study found evidence that TPCs, particularly tactile affective cues, were predominant in affective feedback, compared to general feedback. Firstly, this result extends SSE to the online situation, as users appear to express physical affection to comfort another person, alike to FtF communication. Secondly, this finding further supports the SIP framework, as people adapt their communicative means to the medium, and in highly contextual ways in SNSs (Vandergriff, 2013). Finally, this finding remarkably illustrates the importance of touch for humans, a sense which often acts as an emotion intensifier in FtF communication (Knapp & Hall, 1997), a property similar to that of emoticons (Derks et al., 2007).

7.6. Emotional Mimicry

Since TPC expressions appear to occur contextually in online SSE and previous experimental studies have found correspondence between affective qualities of CMC and FtF (Walther et al., 2005), it may seem surprising that symbols other than tactile cues (for instance auditory ones), were not predictive of their presence in subsequent posts. The first reason to explain this finding may rely on our measurement, since we must note that investigating the possible 'mimicry' of these symbols in its entirety escapes the scope of this investigation. Establishing a causal link between the user and receiver would require an experimental design, in which the correspondence of specific cues would be strictly examined. We believe that this represents a relevant endeavour for future research.

A second reason for this finding may be the exclusion of other factors in our analyses, for instance, the previous existence of online or offline friendships, the duration of the online friendship, belonging to the same gender, sub or main culture. We may speculate that these factors may influence whether nonverbal mimicking takes place, earlier studies have found evidence of language (stylistic) accommodation online between individuals (e.g. Welbers & de Nooy, 2014). Moreover, this accommodation has been found to be influenced by aspects such as belonging to the same group or power aspects in the relationship (Muir, Joinson, Cotterill, & Dewdney, 2016). This idea is further supported by recent research findings of differences in TPC interpretation in friends versus strangers (Riordan & Trichtinger, 2017), which suggests that several particular factors may influence the correspondence of these symbols between the sender and receiver. Further research into online mimicry, or the phenomenon of 'paralinguistic accommodation', considering different contextual, individual and emotional factors in CMC, is recommended.

Lastly, we believe that our finding related to tactile cues, though it may appear modest at first, may bring importance to a larger implication, that is, the issue of enabling effective means of providing affective feedback online for users, for instance by including a button to virtually 'hug' the initiator, an option which still is lacking in several platforms, despite the fact that this reaction may seem more appropriate in the case of sharing highly negative emotions or events, as the popular button 'love' (recently incorporated in the Facebook reactions buttons for instance), may signal that the listener somehow likes or enjoys what is being described. This implication seems overly relevant when considering that users seem to share more positive emotions in SNSs than negative ones (Walther, 2011) and that managing interpersonal impressions in SNSs may be challenging, particularly considering that most users are connected to a multiplicity of different offline contacts online, or the multiple audience problem (Vitak, 2012).

8. Limitations

We note at least three major limitations of our study. The first lies in its observational nature, due to which we cannot be certain about users' actual emotional responses underlying the coding and decoding of TPC in messages. Experimental studies of online SSE in action seem needed to assess the degree to which TPC are actual expressions of emotions.

A second limitation of this research is its parsimonious approach that considers TPC presence as additive in the message for intensity, as it may be argued that certain combinations of TPC may outweigh one another (for instance, ☺ and ☹ when occurring in the same sentence). We must note that, if this was encountered in our sample, it was only very rarely, and posit that while certain symbols may cancel each other out, this may affect the overall message's valence, not its intensity. Additionally and relatedly, our study did not account for other sentiment analysis aspects, such as the presence of sentence intensifiers or modifiers (i.e. very, not), because our focus did not lie on the relationship between the verbal and nonverbal aspects of the message, but focused on the nonverbal aspect. In spite of this, we note that previous studies have found a very low occurrence of these sentence modifiers in their corpora (i.e. Tagliamonte & Denis, 2008).

A third limitation concerns Live Journal's dual nature of a blog SNSs. It may be that the greater anonymity of the platform may have affected the manifestation of online SSE in some aspects, affecting the generalizability of our study to other SNSs. Therefore, we encourage replications in other more popular platforms (i.e. Facebook).

9. Conclusion

Our study brings light into the relational dynamics of online communication within the specific context of online SSE. We find that users seem adept are conveying emotions through TPC while socially sharing emotions online, and that TPC are used in contextual ways within the setting of the interaction. Particularly, haptic cues are an important means by which users may transmit physical affection online. Overall, and in line with other studies (Riordan & Trichtinger, 2017), our study highlights the importance of using a 'contextual lens' to study TPC usage in CMC.

Author disclosure statement

No competing financial interests exist.

Acknowledgements

We would like to reiterate our thanks to Dr. A. Neviarouskaya for providing access to the Affect database, and to M. Vriens for his assistance during the sample generation procedure. We would additionally like to thank the coders for this project.

This research was supported by the Chilean Commission for Scientific and Technological Research (CONICYT) (72140623). The founding entity was not involved in the study design, data collection and analysis, data interpretation, nor the writing and publication process for this article.

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.chb.2017.04.007.

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